



AWS Key Management Service (AWS KMS)

Lecture Slides

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Good luck on your exam!

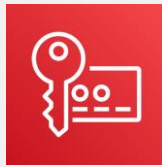
Key Management Service (KMS)



Create, manage, and control **encryption keys**

Compatible with various **AWS services** and **custom apps**

checkbox secure ☒ and start encrypting



Introduction to KMS

KMS makes it easy for you to **create, control and rotate encryption keys** used to encrypt your data on AWS

KMS is a **multi-tenant** Hardware Security Module (HSM)

Most AWS services can just

1. checkbox on Encryption
2. and then choose a KMS key

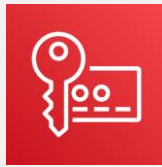
The screenshot shows a portion of the AWS console. A red curved arrow points from the text 'checkbox on Encryption' to the 'Encryption' checkbox, which is checked. Below it, the 'Master Key' dropdown menu is open, showing '(default) aws/ebs' as the selected key. A blue highlight is around the dropdown menu.



KMS can be used with **CloudTrail** to audit access history

KMS integrates with:

- AWS Services:
 - **S3, EC2, RDS, DynamoDB, SQS, SNS, CodeCommit, CodeDeploy, CloudTrail**
 - Other services like **EBS, Lambda, WorkMail, X-Ray, ElastiCache**
- Auditing: Works seamlessly with **AWS CloudTrail** to track key usage and access history



Introduction to KMS

KMS is a **multi-tenant** Hardware Security Module (HSM)

What is Hardware Security Module (HSM)?

- Specialized hardware for storing encryption keys
- Tamper-proof and stores keys in-memory only
- Keys are never written to disk



An HSM server

What is Multi-tenant?

Multi-tenant means **that multiple customers are utilizing the same piece of hardware.**

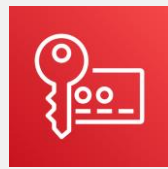
Customers are isolated from each other virtually. If one customer used the entire piece of Hardware (dedicated) that would be called single-tenant



CloudHSM is a single-tenant HSM that gives you full control.

A dedicated HSM means you can meet stricter compliance **FIPS 140-2 Level 3**

KMS is only FIPS Customer master keys are the primary resources in AWS KMS.



KMS – AWS CLI

You can perform many KMS actions through the AWS CLI.

It's worth mentioning some of these API commands since they may appear on the exam:



aws kms create-key

Creates a unique customer-managed customer master key (CMK) in your AWS account and Region



aws kms encrypt

Encrypts plaintext into ciphertext by using a customer master key (CMK)



aws kms decrypt

Decrypts ciphertext that was encrypted by an AWS KMS customer master key (CMK)



aws kms re-encrypt

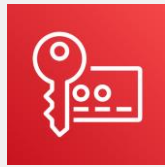
Decrypts ciphertext and then re-encrypts it within AWS KMS.

- manually rotate a CMK
- change the CMK that protects a ciphertext
- change the encryption context of a ciphertext



aws kms enable-key-rotation

Enables automatic rotation of the key material for the specified symmetric customer master key (CMK). You cannot perform this operation on a CMK in a different AWS account.



KMS – Customer Master Key

What is encryption?

The process of encoding a message or information so that only authorized parties can access it and those who are not authorized cannot.

What are cryptographic keys (data key)?

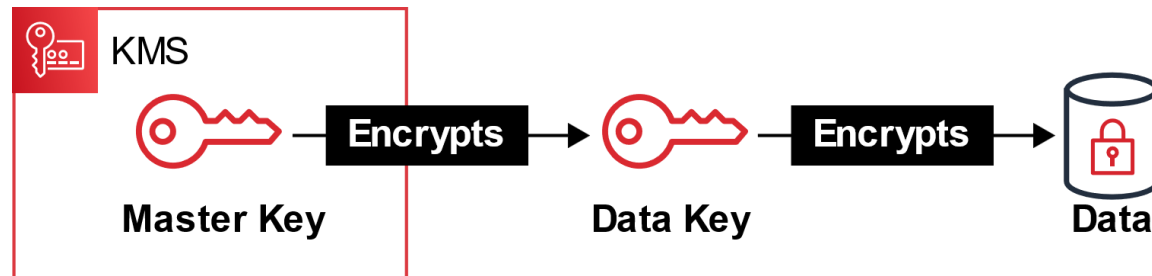
A string of data used to lock or unlock cryptographic functions, including authentication, authorization and encryption.

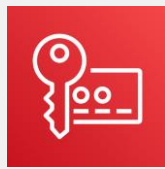
What is a Master Key?

Stored in secure hardware. Master keys are used to encrypt all other keys on a system.

What is Envelope Encryption?

A key used to encrypt another key.





KMS – Customer Master Key

Customer master keys are the primary resources in AWS KMS.

A **customer master key** (CMK) is a **logical representation** of a master key.

The CMK includes metadata, such as:

- the key ID
- creation date
- description
- and key state

The CMK also contains the key material used to:

- encrypt
- and decrypt data.

AWS KMS supports **symmetric** and **asymmetric** CMKs.

Symmetric Key

A 256-bit key that is used for encryption and decryption



Uses **One** Key



Encrypting a S3 bucket using AES-256

Asymmetric Key

an RSA key pair that is used for encryption and decryption or signing and verification (but not both)



Uses **Two** Keys (key pair eg. public and private)



*EC2 key pairs used to SSH into a server



KMS Cheat Sheet

- **Key Management Service (KMS)** creates and manages encryption keys for a variety of AWS Services or your apps.
- KMS can be used with CloudTrail to audit keys' access history.
- KMS can automatically rotate out your keys every year with no need to re-encrypt
- **Customer master keys (CMKs)** are the primary resources in KMS.
- KMS is a multi-tenant HSM, multi-tenant means you are sharing the hardware with multiple customers
- Hardware Security Module (HSM) is a specialized hardware for storing your keys and is tamper-proof
- **KMS** is up to **FIPS 140-2 Level 2** compliant
- KMS stores Master Keys (not data keys)
- Master Keys are used to encrypt data keys which is called **Envelope Encryption**
- KMS supports two types of keys **symmetric** and **asymmetric**
 - **Symmetric** is a single key using 256-bit encryption eg. S3 bucket AES-256
 - **Asymmetric** uses two keys eg. key Pair public and private
- Important KMS APIs to remember:
 - **aws kms create-key** – creates a key
 - **aws kms encrypt** – encrypts a key
 - **aws kms decrypt** – decrypts a key
 - **aws kms re-encrypt** – re-encrypts a key
 - **aws kms enable-key-rotation** – turn on automatic key rotation (only for **symmetric keys**)